Pariial Differential Equations – Exercises

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Contents

Lecture 1: Fourier Theory

29. August 2025

Exercise 1.

Consider the following periodic functions. Find possible values for the period, using the definition of periodic functions.

(a) $f(x) = e^{\cos x}$

(b) $f(x) = \cos\left(x + \frac{\pi}{3}\right)$

(c) $f(x) = \cos \frac{\pi x}{3}$

(d) $f(x) = \cos x + \cos 3x$

(e) $f(x) = \cos x \cos 3x$

(f) $f(x) = \cos x + \cos(0{,}6x)$

For (a) the exponent to e, i.e. $\cos x$ only has a range of $(-1; 1)$. This means that the exponent will change between -1 and 1 with a period of 2π, therefore the period $p_a = 2\pi$ will also be the case for the function described in (a).

For (b) we are once again working with a cos function. This time it is however shifted by $\frac{\pi}{3}$ along the x-axis. A translation along the x-axis will, however, not affect the period of the function and therefore the period here is once again $p_b = 2\pi$.

For (c) the x is multiplied by $\frac{\pi}{3} \approx 1{,}047$. This means that the argument to the cosine will reach 2π at a speed that is $\frac{\pi}{3} \approx 1{,}047$ times faster than a normal cosine. Therefore the period of this is $p_c = \frac{3 \cdot 2\pi}{\pi} = 6$

For (d) we use that

$$f(x) = f(x + p)$$

for a periodic function to get

$$\cos x + \cos 3x = \cos(x+p) + \cos(3(x+p))$$
$$\cos x + \cos 3x = \cos(x+p) + \cos(3x+3p)\,.$$

We see that this is true when both p and $3p$ are integer multiples of 2π so $p_d = 2\pi$

For (e) we use the same definition to get

$$\cos x \cos 3x = \cos(x+p)\cos(3x+3p)\,.$$

We see that this is also true only when both p and $3p$ are integer multiples of 2π so $p_e = 2\pi$

For (f) we can once again use the same definition to get

$$\cos x + \cos(0{,}6x) = \cos(x+p) + \cos(0{,}6x + 0{,}6p)\,.$$

This is true when p and $0{,}6p$ are integer multiples of 2π, so $p_f = 10\pi$.

Exercise 2.

Let

$$f(x) = x, \quad -1 \le x \le 1, \quad f(x) = f(x+p), \quad p = 2.$$

- Sketch f.
- Find a Fourier series representation of f.

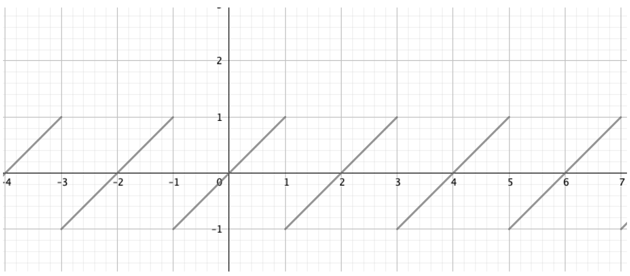

Figure 1:

f has been sketched on **Figure 1**. The Fourier series of this can, as it is a "nice" function be found by the formula:

$$f(x) = a_0 + \sum_{n=1}^{\infty}\left(a_n \cos\frac{n\pi x}{L} + b_n \sin\frac{n\pi x}{L}\right).$$

Where $L = \frac{p}{2} = 1$ with $p = 2$ being the period of the function. The Fourier coefficients can be found by the Euler formulas:

$$a_0 = \frac{1}{2L}\int_{-L}^{L} f(x)\,\mathrm{d}x$$
$$a_n = \frac{1}{L}\int_{-L}^{L} f(x)\cos\frac{n\pi x}{L}\,\mathrm{d}x, \quad n = 1,2,3,\ldots$$
$$b_n = \frac{1}{L}\int_{-L}^{L} f(x)\sin\frac{n\pi x}{L}\,\mathrm{d}x, \quad n = 1,2,3,\ldots.$$

We start by solving for a_0 as:

$$a_0 = \int_{-1}^{1} x \,\mathrm{d}x = \left[\frac{1}{2}x^2\right]_{-1}^{1} = \cdot 0 = 0.$$

We now solve for a_n using integration by parts as:

$$\begin{aligned}
a_n &= \int_{-1}^{1} x \cos n\pi x \,\mathrm{d}x \\
&= \left[x \frac{\sin n\pi x}{n\pi}\right]_{-1}^{1} - \int_{-1}^{1} \frac{\sin n\pi x}{n\pi} \,\mathrm{d}x \\
&= \left(-\frac{1}{n\pi}\right) \int_{-1}^{1} \sin n\pi x \,\mathrm{d}x \\
&= \left[-\frac{1}{n\pi}\right] \left[-\frac{\cos n\pi x}{n\pi}\right]_{-1}^{1} \\
&= 0.
\end{aligned}$$

We similarly solve for b_n as:

$$\begin{aligned}
b_n &= \int_{-1}^{1} x \sin n\pi x \,\mathrm{d}x \\
&= \left[x \frac{-\cos n\pi x}{n\pi}\right]_{-1}^{1} - \int_{-1}^{1} \frac{-\cos n\pi x}{n\pi} \,\mathrm{d}x \\
&= \left(-\frac{2}{n\pi}\right) \cos n\pi + \frac{1}{n\pi} \left[\frac{\sin n\pi x}{n\pi}\right]_{-1}^{1} \\
&= \left(-\frac{2}{n\pi}\right) \cos n\pi.
\end{aligned}$$

The full Fourier series therefore becomes:

$$f(x) = \sum_{n=1}^{\infty} \left(-\frac{2}{n\pi}\right) \cos n\pi \cdot \sin n\pi x.$$

Exercise 3.

Let $a \neq 0$ and $b \neq 0$. Let $f(x) = f(x + p_0)$ be a periodic function with period p_0. Is $f(ax + b)$ periodic? If yes, find a period. Give arguments for your answer.

The b only serves to translate the function $f(x)$ along the x-axis, this does not change the period. The value of a corresponds to how "quickly" some value is reached, e.g. if ax is an argument to a function a doubling of the value of a will mean that the function will reach its maximum for a value of x only half as big. I.e. there is an inverse relation between the size of a and the period as:

$$p_2 = \frac{p_1}{a}.$$

Exercise 4.

Calculate the left hand limit and the right hand limit of

$$f(x) = \frac{|x|}{x}$$

at $x = 0$.

We start with the right-hand limit as:

$$f(0+) = \lim_{h\to 0} (0+h) = \lim_{h\to 0} \frac{|h|}{h} = \lim_{h\to 0} \frac{h}{h} = 1.$$

And now we can similarly do the left-hand limit as:

$$f(0-) = \lim_{h\to 0} (0-h) = \lim_{h\to 0} \frac{|-h|}{-h} = \lim_{h\to 0} \frac{h}{-h} = -1.$$

Exercise 5.

Calculate the left-hand derivative and the right-hand derivative of

$$f(x) = x\,|x|$$

at $x = 0$.

We start with the right-hand derivative as:

$$f'(0+) = \lim_{h\to 0} \frac{f(0+h) - f(0)}{h} = \lim_{h\to 0} \frac{h|h|}{h} = \lim_{h\to 0} h = 0.$$

And the left-hand derivative can likewise be computed as:

$$f'(0-) = \lim_{h\to 0} \frac{f(0) - f(0-h)}{h} = \lim_{h\to 0} \frac{-(-h)|-h|}{h} = \lim_{h\to 0} h = 0.$$

Lecture 2: Even and odd functions

5. September 2025

Exercise 1.

Find a period and the corresponding Fourier coefficients of

a) $f(x) = \sin\left(\frac{\pi}{4}\right)\sin\left(\frac{2x}{4}\right) + \cos\left(\frac{\pi}{5}\right)\sin\left(\frac{3x}{7}\right)$

We rewrite the given function as:

$$\begin{aligned} f(x) &= \sin\left(\frac{\pi}{4}\right)\sin\left(\frac{2x}{4}\right) + \cos\left(\frac{\pi}{5}\right)\sin\left(\frac{3x}{7}\right) \\ &= \sin\left(\frac{\pi}{4}\right)\sin\left(\frac{7\pi x}{14\pi}\right) + \cos\left(\frac{\pi}{5}\right)\sin\left(\frac{6\pi x}{14\pi}\right). \end{aligned}$$

We will now compare this to the general formulation of the Fourier series as:

$$f(x) = a_0 + \sum_{n=1}^{\infty} \left(a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right)\right).$$

We can quickly see that $L = 14\pi \implies p = 28\pi$, $b_6 = \cos\left(\frac{\pi}{5}\right)$, $b_7 = \sin\left(\frac{\pi}{4}\right)$. All other unmentioned Fourier coefficients are zero.

b) $f(x) = \sin\left(\sqrt{2}x\right) + \cos\left(\frac{x}{\sqrt{2}}\right)$

We once again start by rewriting as:

$$\begin{aligned} f(x) &= \sin\left(\sqrt{2}x\right) + \cos\left(\frac{x}{\sqrt{2}}\right) \\ &= \sin\left(\frac{2\pi x}{\sqrt{2}\pi}\right) + \cos\left(\frac{\pi x}{\sqrt{2}\pi}\right). \end{aligned}$$

Comparing this with the general formulation of the Fourier series:

$$f(x) = a_0 + \sum_{n=1}^{\infty}\left(a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right)\right).$$

We see that $L = \sqrt{2}\pi \implies p = 2\sqrt{2}\pi$, $a_1 = 1$, $b_2 = 1$. All other Fourier coefficients are zero.

c) $f(x) = \sin\left(\frac{\pi}{3} + \frac{3\pi x}{4}\right)$

We use that

$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \sin\beta\cos\alpha$$

to rewrite the equation as:

$$\begin{aligned} f(x) &= \sin\left(\frac{\pi}{3} + \frac{3\pi x}{4}\right) \\ &= \sin\frac{\pi}{3}\cdot\cos\frac{3\pi x}{4} + \sin\frac{3\pi x}{4}\cdot\cos\frac{\pi}{3}. \end{aligned}$$

Comparing this to the general formulation of the Fourier series

$$f(x) = a_0 + \sum_{n=1}^{\infty}\left(a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right)\right)$$

we see that $L = 4 \implies p = 8$, $a_3 = \sin\frac{\pi}{3}$, $b_3 = \cos\frac{\pi}{3}$. All other Fourier coefficients are zero

Exercise 2.

Are the following functions even or odd? Give arguments.

a) $f(x) = \left(\cos x^2 + \cos x\right)\sin x$

We check if the function is even or odd by checking what happens for an input of $-x$ as:

$$\begin{aligned} f(-x) &= \left(\cos(-x)^2 + \cos(-x)\right)\sin(-x) \\ &= \left(\cos x^2 + \cos(-x)\right)(-\sin x) \\ &= -f(x). \end{aligned}$$

Hence the function is odd.

b) $f(x) = \frac{1+\cos x}{2+\cos x^2}$

We follow the same procedure:

$$\begin{aligned} f(-x) &= \frac{1+\cos(-x)}{2+\cos(-x)^2} \\ &= \frac{1+\cos x}{2+\cos x} \\ &= f(x). \end{aligned}$$

Hence the function is even.

c) $f(x) = \frac{1+\sin x}{2+\sin x^2}$

The same procedure is applied:

$$\begin{aligned} f(-x) &= \frac{1+\sin(-x)}{2+\sin(-x)^2} \\ &= \frac{1-\sin x}{2+\sin x} \\ &\neq \begin{cases} f(x) \\ -f(x) \end{cases} \end{aligned}$$

Hence the function is neither even nor odd.

d) $f(x) = x|x|$

The same procedure is applied:

$$\begin{aligned} f(-x) &= (-x)\,|-x| \\ &= -x\,|x| \\ &= -f(x). \end{aligned}$$

Hence the function is odd.

Exercise 3.

Consider the periodic function

$$f(x) = \begin{cases} x, & -2 < x < -1 \\ x+k, & -1 < x < 1 \\ x, & 1 < x < 2 \end{cases}$$
$$f(x) = f(x+p), p = 4$$

where k is a constant. Decompose $f(x)$ into its even and odd part.

We apply the definition of decomposition into odd and even parts for the first interval, $-2 < x < -1$ as:

$$f_1(x) = \frac{1}{2}(f(x) + f(-x)) = \frac{1}{2}(x + (-x)) = 0$$
$$f_2(x) = \frac{1}{2}(f(x) - f(-x) = \frac{1}{2}(x - (-x)) = x.$$

For the second interval, $-1 < x < 1$ we get

$$f_1(x) = \frac{1}{2}(f(x) + f(-x)) = \frac{1}{2}((x + k) + (-x + k)) = k$$
$$f_2(x) = \frac{1}{2}(f(x) - f(-x) = \frac{1}{2}((x + k) - (-x + k)) = x.$$

And for the last interval, $1 < x < 2$ we get:

$$f_1(x) = \frac{1}{2}(f(x) + f(-x)) = \frac{1}{2}(x + (-x)) = 0$$
$$f_2(x) = \frac{1}{2}(f(x) - f(-x) = \frac{1}{2}(x - (-x)) = x.$$

We thus have:

$$\begin{aligned} f(x) &= f_1(x) + f_2(x) \\ f_1(x) &= \begin{cases} 0 & -2 < x < -1 \\ k & -1 < x < 1 \\ 0 & 1 < x < 2. \end{cases} \\ f_1(x) &= f_1(x + p), \quad p = 4 \\ f_2(x) &= x, \qquad -2 < x < 2 \\ f_2(x) &= f_2(x + p), \quad p = 4. \end{aligned}$$

Exercise 4.

Consider the periodic function

$$f(x) = \begin{cases} 0, & -\pi < x < 0 \\ 2x, & 0 < x < \pi \end{cases}$$
$$f(x) = f(x + p), p = 2\pi.$$

Decompose $f(x)$ into its even and odd part and find its Fourier series.

Hint: You may want to use examples from the lecture.

For the interval $-\pi < x < 0$ we have

$$f_1(x) = \frac{1}{2}(0 + (-2x)) = -x$$
$$f_2(x) = \frac{1}{2}(0 - (-2x)) = x.$$

And for $0 < x < \pi$ we have

$$\begin{aligned} f_1(x) &= \frac{1}{2}(2x+0) = x \\ f_2(x) &= \frac{1}{2}(2x-0) = x. \end{aligned}$$

We thus have

$$\begin{aligned} f(x) &= f_1(x) + f_2(x) \\ f_1(x) &= |x|, -\pi < x < \pi \\ f_1(x) &= f_1(x+p), \quad p = 2\pi \\ f_2(x) &= x, \quad -\pi < x < \pi \\ f_2(x) &= f_2(x+p) \quad p = 2\pi. \end{aligned}$$

From the lecture we know that f_1 has the Fourier cosine series

$$f_1(x) = \frac{\pi}{2} + \sum_{n=1}^{\infty} \frac{2}{\pi n^2} (\cos n\pi - 1) \cos nx$$

and that f_2 has the Fourier sine series:

$$f_2(x) = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{2}{n} \sin nx.$$

Combining these we get the Fourier series for $f(x)$ as:

$$f(x) = \frac{\pi}{2} + \sum_{n=1}^{\infty} \left(\frac{2}{\pi n^2}(\cos n\pi - 1)\cos nx + (-1)^{n+1} \frac{2}{n} \sin nx \right).$$

Exercise 5.

Decompose

$$f(x) = e^{ix}$$

into its even and odd part.

Hint; you may want to use the representation of $f(x)$ in terms of $\cos x$ and $\sin x$.

We apply the same procedure as in the last couple of exercises whilst remembering that:

$$e^{ix} = \cos x + i \sin x.$$

We therefore get:

$$\begin{aligned} f_1(x) &= \frac{1}{2}(f(x) + f(-x)) \\ &= \frac{1}{2}(\cos x + i\sin x + \cos x - i \sin x) \\ &= \cos x \\ f_2(x) &= \frac{1}{2}(f(x) - f(-x)) \\ &= \frac{1}{2}(\cos x + i \sin x - (\cos x - i \sin x)) \\ &= i \sin x. \end{aligned}$$

This also is what would be expected when looking at the representation of e^{ix} introduced at the start of this answer.

Lecture 3: Fourier Series

12. September 2025

Exercise 1.

Consider the function $f(x) = \cos x, 0 < x < \pi$.

Hint: In some cases the expansion can be an even/odd function.

Hint: For $m, n \in \mathbb{N}, m \neq n, L > 0$:

$$\int_0^{\frac{\pi}{2}} \cos x \sin(2nx)\,\mathrm{d}x = -\frac{2n}{1-4n^2}$$
$$\int_0^L \cos\left(\frac{m\pi x}{L}\right)\sin\left(\frac{n\pi x}{L}\right)\,\mathrm{d}x = \frac{nL}{\pi\left(m^2-n^2\right)}\left(\cos\left(n\pi\right)\cos(m\pi)-1\right)$$
$$\int_0^L \cos\left(\frac{\pi x}{L}\right)\sin\left(\frac{\pi x}{L}\right)\,\mathrm{d}x = 0.$$

(a) Sketch the expansion of f and find the corresponding Fourier series.

The expansion of f is simply the "repetition" of f indefinitely along the x-axis. This is shown on Figure 2

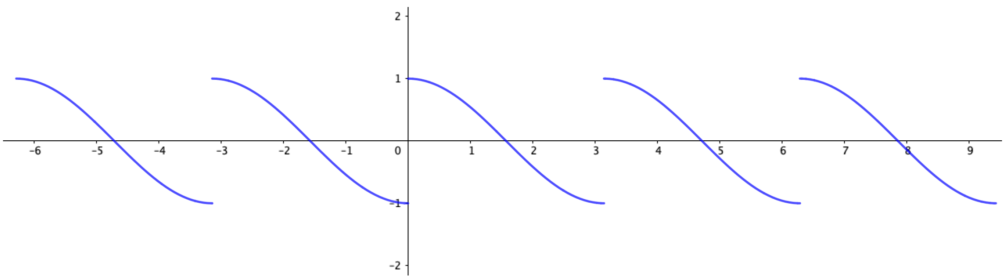

Figure 2: Expansion of $f(x) = \cos x, 0 < x < \pi$

To find the Fourier series of this we must first define a periodic function $f_0(x)$ around $x = 0$ based on Figure 2 we see that this can be written as

$$f_0(x) = \begin{cases} \cos x, & 0 < x < \frac{\pi}{2} \\ -\cos x, & -\frac{\pi}{2} < x < 0. \end{cases}$$
$$f_0(x) = f_0(x+p), p = 2L = \pi.$$

As $f_0(x)$ is an odd function its Fourier series will be a so-called Fourier sine series:

$$f(x) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right).$$

Where the Fourier coefficient b_n is

$$b_n = \frac{2}{L}\int_0^L f(x)\sin\left(\frac{n\pi x}{L}\right)\,\mathrm{d}x.$$

Substituting the general $f(x)$ for our $f_0(x)$ we get:

$$\begin{aligned} b_n &= \frac{4}{\pi}\int_0^{\frac{\pi}{2}} \cos x \sin(2nx)\,\mathrm{d}x \\ &= \frac{4}{\pi}\cdot\left(-\frac{2n}{1-4n^2}\right) \\ &= \frac{8n}{\pi\left(4n^2-1\right)}. \end{aligned}$$

Hence the Fourier series becomes:

$$f_0(x) = \sum_{n=1}^{\infty} \frac{8n}{\pi\left(4n^2-1\right)}\sin(2nx).$$

(b) Sketch the even expansion of f and find the corresponding Fourier cosine series.

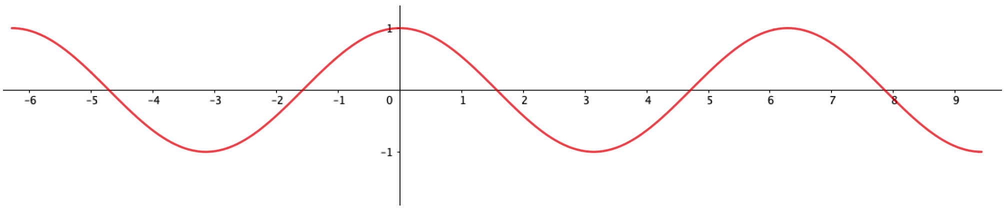

Figure 3: Even expansion of $f(x) = \cos x, 0 < x < \pi$

On Figure 3 it can be seen that the even expansion of the expression is simply the cos function. We can therefore write our periodic function as

$$\begin{aligned} f_1(x) &= \cos x, \quad 0 < x < \frac{\pi}{2} \\ f_1(x) &= f_1(x+p), \quad p = 2L = 2\pi. \end{aligned}$$

This can also be written as

$$f_1(x) = \cos x.$$

We remember that the general form of a Fourier cosine series is

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right)$$

or with the values given in the exercise

$$f_1(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{\pi}\right) = a_0 + \sum_{n=1}^{\infty} a_n \cos nx.$$

Now we can compare these

$$\begin{aligned} \cos x &= a_0 + \sum_{n=1}^{\infty} a_n \cos nx \\ &\implies a_0 = 0, a_1 = 1. \end{aligned}$$

(c) Sketch the odd expansion of f and find the corresponding Fourier sine series. The odd expansion gives

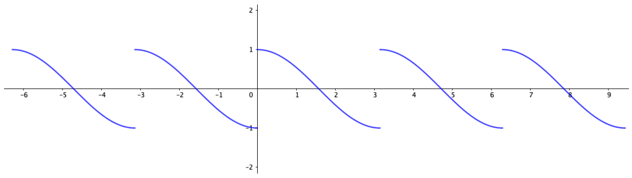

Figure 4: Odd expansion of $f(x) = \cos x, 0 < x < \pi$

the periodic function

$$f_2(x) = \begin{cases} \cos x, & 0 < x < \pi \\ -\cos x & -\pi < x < 0. \end{cases}$$

$$f_2(x) = f_2(x+p), \quad p = 2L = 2\pi.$$

We remember that the general form of a Fourier sine series is

$$f(x) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right)$$

or with the values in the exercise

$$f_2(x) = \sum_{n=1}^{\infty} b_n \sin(nx)$$

where b_n is

$$b_n = \frac{2}{L}\int_0^L f(x)\sin\left(\frac{n\pi x}{L}\right)\,\mathrm{d}x$$

or

$$b_n = \frac{2}{\pi}\int_0^{\pi} \cos x \sin(nx)\,\mathrm{d}x.$$

For $n = 1$ this simplifies to

$$b_1 = 0.$$

For $n > 1$ it becomes

$$\begin{aligned} b_n &= \frac{2}{\pi}\cdot\frac{n\pi}{\pi(1-n^2)}\left(-\cos(n\pi) - 1\right) \\ &= \frac{2n}{\pi(n^2-1)}\left(1 - \cos(n\pi)\right). \end{aligned}$$

Hence the full Fourier sine series is

$$f_2(x) = \sum_{n=2}^{\infty} \frac{2n}{\pi(n^2-1)}\left(1 + \cos(n\pi)\right)\sin(nx).$$

(d) Calculate the error E_N of the expansion and the odd expansion for $N = 1, 2, 3$. What can be said about the quality of the approximations for these two expansions?

The general formula for calculating the error E_N of the approximation is

$$E_N = \int_{-L}^{L} f^2(x)\,\mathrm{d}x - L\left(2a_0^2 + \sum_{n=1}^{N}\left(a_n^2 + b_n^2\right)\right).$$

If we start by considering the *expansion* we are working with the expression

$$f_0(x) = \sum_{n=1}^{\infty} \frac{8n}{\pi\left(4n^2-1\right)} \sin\left(2nx\right).$$

Over the entire interval $[-L;L]$ $f_0^2 = \cos^2$. Using this fact and substituting $L = \frac{\pi}{2}$, $a_n = 0$ and $b_n = \frac{8n}{\pi(4n^2-1)}$ we can write

$$\begin{aligned}
E_N &= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^2(x)\,\mathrm{d}x - \frac{\pi}{2}\sum_{n=1}^{N}\left(\frac{8n}{\pi\left(4n^2-1\right)}\right)^2 \\
&= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1}{2}\left(1+\cos(2x)\right)\,\mathrm{d}x - \frac{\pi}{2}\sum_{n=1}^{N}\frac{64n^2}{\pi^2\left(4n^2-1\right)^2} \\
&= \frac{\pi}{2} + \frac{1}{2}\left[\frac{1}{2}\sin\left(2x\right)\right]_{-\frac{\pi}{2}}^{\frac{\pi}{2}} - \frac{32}{\pi}\sum_{n=1}^{N}\frac{n^2}{\left(4n^2-1\right)^2} \\
&= \frac{\pi}{2} - \frac{32}{\pi}\sum_{n=1}^{N}\frac{n^2}{\left(4n^2-1\right)^2}.
\end{aligned}$$

Calculating the numerical value of this for $n = 1,2,3$ yields

$$\begin{aligned}
E_1 &= \frac{\pi}{2} - \frac{32}{9\pi} \approx 0{,}439 \\
E_2 &= \frac{\pi}{2} - \frac{32}{\pi}\left(\frac{1}{9} + \frac{4}{15^2}\right) \approx 0{,}258 \\
E_3 &= \frac{\pi}{2} - \frac{32}{\pi}\left(\frac{1}{9} + \frac{4}{15^2} + \frac{9}{35^2}\right) \approx 0{,}183.
\end{aligned}$$

Now moving on to the odd expansion our $f_2^2(x) = \cos^2(x)$ as before, however this time $L = \pi$ and $b_n = \frac{2n}{\pi(n^2-1)}\left(1+\cos\left(n\pi\right)\right)$. Plugging these values into the formula we get:

$$\begin{aligned}
E_N &= \int_{-\pi}^{\pi} \cos^2(x)\,\mathrm{d}x - \pi\sum_{n=2}^{N}\left(\frac{2n}{\pi\left(n^2-1\right)}\left(1+\cos\left(n\pi\right)\right)\right)^2 \\
&= \pi - \pi\sum_{n=2}^{N}\frac{4n^2}{\pi^2\left(n^2-1\right)^2}\left(1+\cos\left(n\pi\right)\right)^2 \\
&= \pi - \frac{4}{\pi}\sum_{n=2}^{N}\frac{n^2}{\left(n^2-1\right)^2}\left(1+\cos(n\pi)\right)^2.
\end{aligned}$$

And for $n = 1,2,3$ we get

$$\begin{aligned}
E_1 &= \pi - \frac{4}{\pi}\cdot 0 = \pi \\
E_2 &= \pi - \frac{4}{\pi}\left(0 + \frac{16}{9}\right) \approx 0{,}878 \\
E_3 &= \pi - \frac{4}{\pi}\left(0 + \frac{16}{9} + \frac{9}{64}\cdot 0\right) \approx 0{,}878.
\end{aligned}$$

When comparing errors we must take into account the length of the intervals L. Therefore a realistic comparison requires us to half the errors from the odd expansion. Even when doing so the error rate from the "normal" expansion performs better and should therefore be preferred.

Exercise 2.

Find the Fourier integral representation of

$$f(x) = \begin{cases} 1, & -1 < x < 1 \\ -1, & 1 < x < 2 \\ -1, & -2 < x < -1 \\ 0, & \text{otherwise.} \end{cases}.$$

The Fourier integral is given by

$$f(x) = \int_0^\infty \left(A(\omega)\cos(\omega x) + B(\omega)\sin(\omega x)\right) \,\mathrm{d}\omega$$

where

$$A(\omega) = \frac{1}{\pi}\int_{-\infty}^{\infty} f(x)\cos(\omega x)\,\mathrm{d}x$$

and

$$B(\omega) = \frac{1}{\pi}\int_{-\infty}^{\infty} f(x)\sin(\omega x)\,\mathrm{d}x.$$

We start by finding $A(\omega)$ using piecewise integration

$$\begin{aligned} A(\omega) &= \frac{1}{\pi}\left(\int_{-2}^{-1} -\cos(\omega x)\,\mathrm{d}x + \int_{-1}^{1}\cos(\omega x)\,\mathrm{d}x + \int_{1}^{2} -\cos(\omega x)\,\mathrm{d}x\right) \\ &= \frac{4\sin\omega - 2\sin(2\omega)}{\pi\omega}. \end{aligned}$$

The same technique can now be applied to find $B(\omega)$ as

$$\begin{aligned} B(\omega) &= \frac{1}{\pi}\left(\int_{-2}^{-1} -\sin(\omega x)\,\mathrm{d}x + \int_{-1}^{1}\sin(\omega x)\,\mathrm{d}x + \int_{1}^{2} -\sin(\omega x)\,\mathrm{d}x\right) \\ &= 0. \end{aligned}$$

Therefore the Fourier integral representation of $f(x)$ is

$$f(x) = \int_0^\infty \left(\frac{4\sin\omega - 2\sin(2\omega)}{\pi\omega}\right)\cos(\omega x)\,\mathrm{d}\omega.$$

Exercise 3.

Find the Fourier integral representation of

$$f(x) = \begin{cases} -1, & -2 < x < 0 \\ 1, & 0 < x < 2 \\ 0, & \text{otherwise.} \end{cases}.$$

Here the same procedure can be applied as above to get

$$\begin{aligned} A(\omega) &= \frac{1}{\pi}\left(\int_{-2}^{0} -\cos(\omega x)\,\mathrm{d}x + \int_{0}^{2} \cos(\omega x)\,\mathrm{d}x\right) \\ &= 0. \end{aligned}$$

and

$$\begin{aligned} B(\omega) &= \frac{1}{\pi}\left(\int_{-2}^{0} -\sin(\omega x)\,\mathrm{d}x + \int_{0}^{2} \sin(\omega x)\,\mathrm{d}x\right) \\ &= \frac{2-2\cos(2\omega)}{\pi\omega}. \end{aligned}$$

Which becomes

$$f(x) = \int_{0}^{\infty} \frac{2-2\cos(2\omega)}{\pi\omega}\sin(\omega x)\,\mathrm{d}\omega.$$

Exercise 4.

Consider the periodic rectangular wave and its Fourier series as discussed in Lecture 1. Calculate the error of the N-th approximation for $N = 1, 2, 3$.

The general formula for calculating the error E_N of the approximation is

$$E_N = \int_{-L}^{L} f^2(x)\,\mathrm{d}x - L\left(2a_0^2 + \sum_{n=1}^{N}\left(a_n^2 + b_n^2\right)\right).$$

In this case $f(x)^2 = k^2$, $L = \pi$, $a_0 = 0$, $a_n = 0$, and $b_n = \frac{2k}{n\pi}(1-\cos(n\pi))$. I.e.

$$\begin{aligned} E_N &= \int_{-\pi}^{\pi} k^2\,\mathrm{d}x - \pi\sum_{n=1}^{\infty}\left(\frac{2k}{n\pi}\left(1-\cos\left(n\pi\right)\right)\right)^2 \\ &= 2\pi k^2 - \frac{4k^2}{\pi}\sum_{n=1}^{N}\frac{1}{n^2}\left(1-\cos(n\pi)\right)^2. \end{aligned}$$

For $n = 1, 2, 3$ we get $E_1 = E_2 \approx 1{,}19k^2$ and $E_3 \approx 0{,}62k^2$.